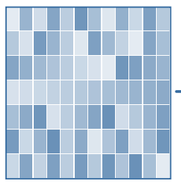
GCN-conditioned count forecasting with Flow Matching

H13 | next 10 minutes | fixed inputs

(a) Graph-conditioned encoder

Weekly profiles



$P \in \mathbb{R}^{N \times 420}$

Weekly MLP
$420 \to 256 \to 64$

Z

Fixed OD graph A



$\bar{Z} = AZ$

Existing context
priors / calendar / phase
weather / recent net + masks

Context fusion
$164, 72 \to 256$

$h_i \in \mathbb{R}^{256}$

Weekly MLP + fixed OD aggregation + context MLP

Full encoder layers are expanded in Fig. 2(a).

(b) Count decoders

Poisson likelihood during count training

Rate head
$256 \to 2$

λ_i

rental / return

Poisson reference
$Y_R, Y_D \sim \text{Pois}(\lambda_i)$

$\lambda = p \exp(2\tanh(a/2))$

Separate sampling route

$\mu_i = \log(1 + \lambda_i)$

Conditional Flow Matching

Gaussian noise



$r_0 \sim \mathcal{N}(0, I_2)$

h_i

$v_\theta(r_s, s \mid h_i, \mu_i)$

$264 \to 128 \to 128 \to 128 \to 2$

r_1

log residual

ODE sampling: 16 Heun steps

128 IID particles across seeds 11 and 29

$\widehat{Y} = \max(0, \lfloor \exp(r_1 + \mu_i) - 1 \rfloor)$

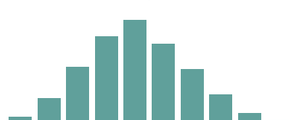
128×2 counts

counts

(c) Inventory projection

Current inventory
$b_{i,t}$ (API observation)

Inventory balance
$b_{i,t+10} = \max(0, b_{i,t} + Y_D - Y_R)$



Median / interval / P(stock = 0)

Inventory distribution (128 particles)

Dispatch optimizer
pickup / delivery quantities and route

TRAINING

Seoul count pretraining
initialize weekly encoder

Tashu historical count training
fit encoder + rate head / Poisson NLL

Freeze count network; train FM
residual velocity matching / MSE

N: stations. R: rentals. D: returns. Dashed arrows: conditioning. Distribution glyphs are schematic.